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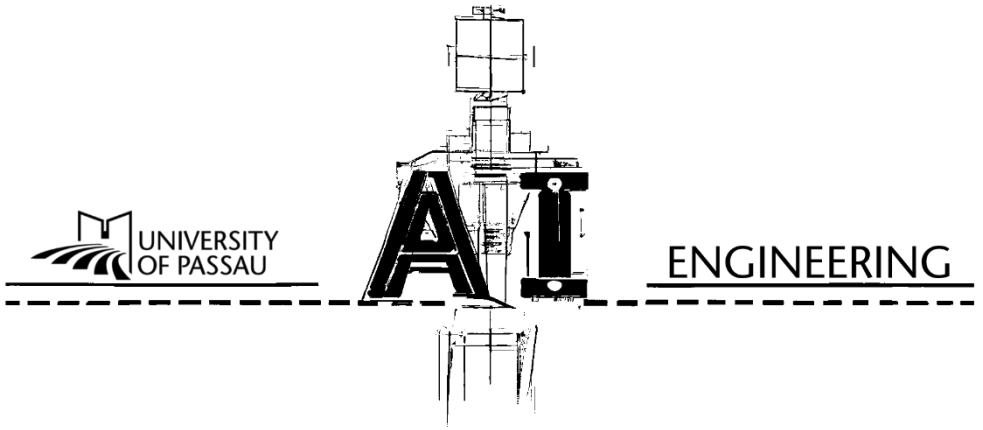
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Understanding or Memorizing? A Case Study of German Definite Articles in Language Models

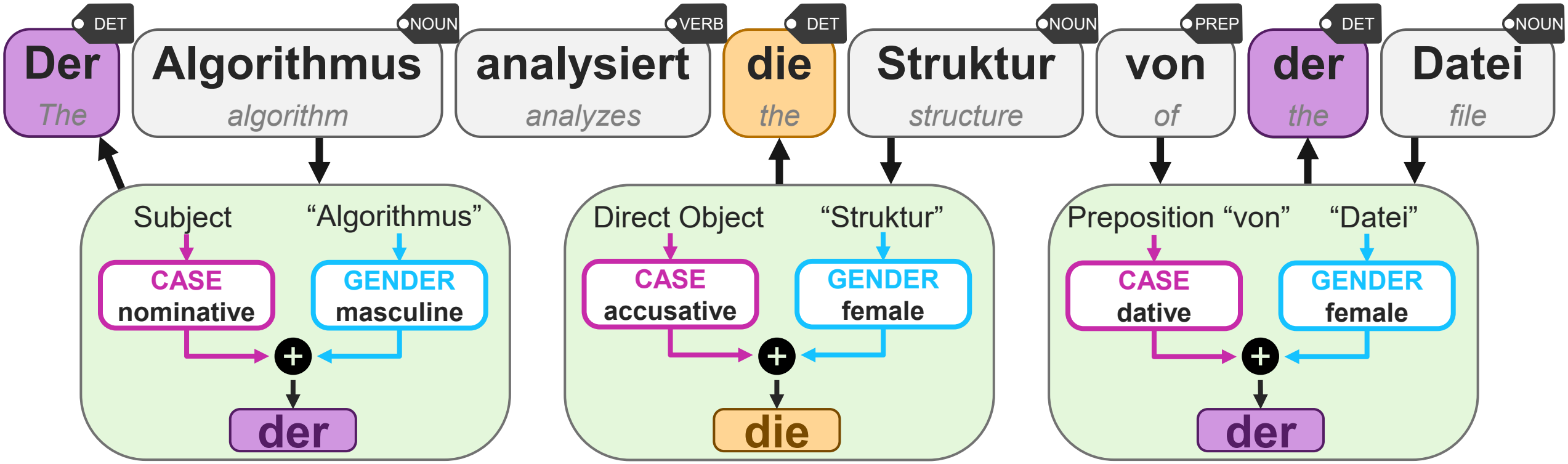
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Singular German Articles Depend on Case and Gender



	Nominative	Accusative	Dative	Genitive
Masculine	der	den	dem	des
Female	die	die	der	der
Neutral	das	das	dem	des

6 Articles
Highly syncretic
The same surface form occurs in multiple gender-case cells

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Correct Predictions Do Not Reveal The Mechanism

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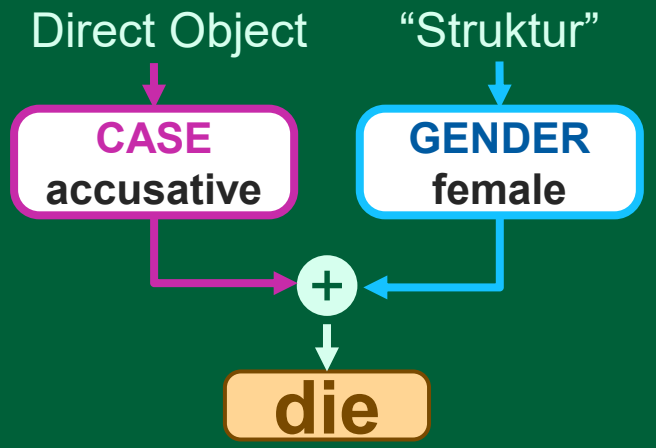
Der	Algorithmus	analysiert	[MASK]	Struktur	von	der	Datei
The	*algorithm*	*analyzes*		*structure*	*of*	*the*	*file*

Language Model

die	0.93
der	0.04
das	0.02

High-confidence prediction

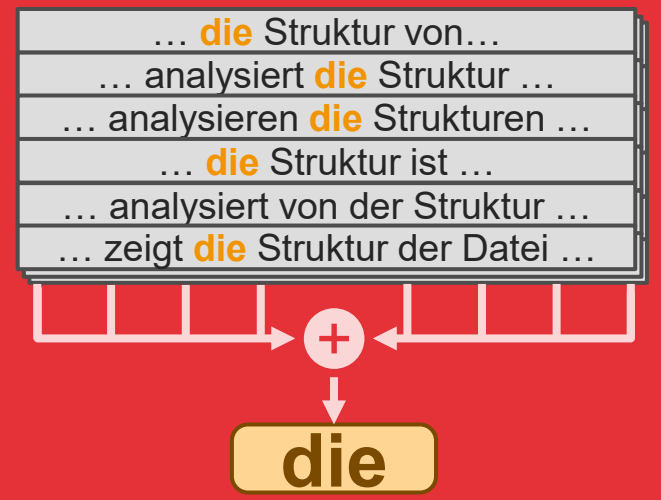
Rule-Encoding



abstract grammatical features

? But Why ?

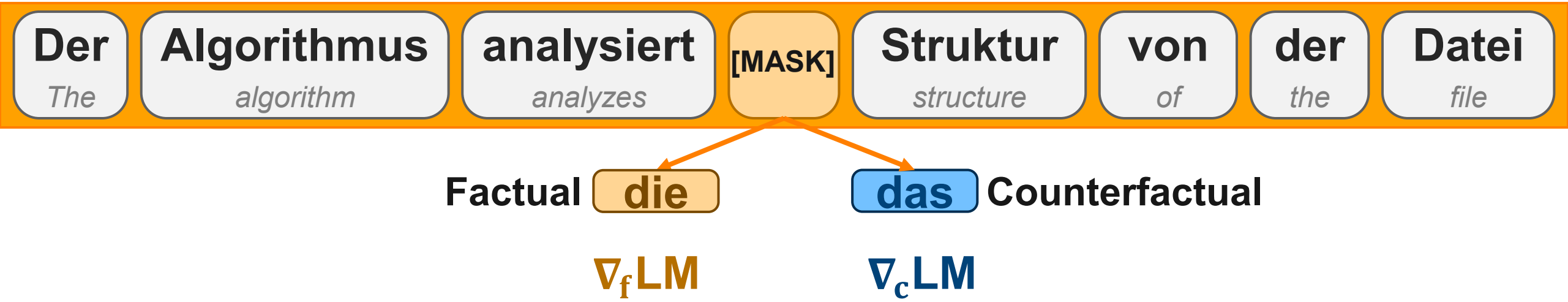
Memorization



surface pattern recall

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Factual vs. Counterfactual Article Targets



$\nabla_{\cdot}$LM denotes the MLM gradient derived using the respective target token

From Article Pairs to Gender-Case Transitions

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Der	Algorithmus	analysiert	[MASK]	Struktur	von	der	Datei
The	*algorithm*	*analyzes*		*structure*	*of*	*the*	*file*

Die	Analyse	erklärt	[MASK]	Verhalten	beim	Training
The	*analysis*	*explains*		*behavior*	*during*	*training*

Gender-Case Transition: (NEUT, ACC) ↔ (FEM, ACC)

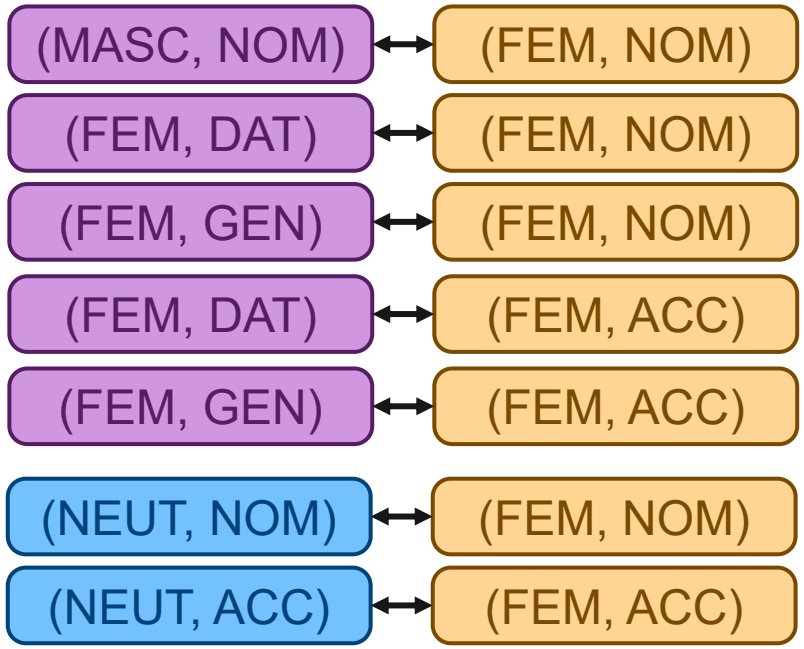
Surface Realization: das ↔ die

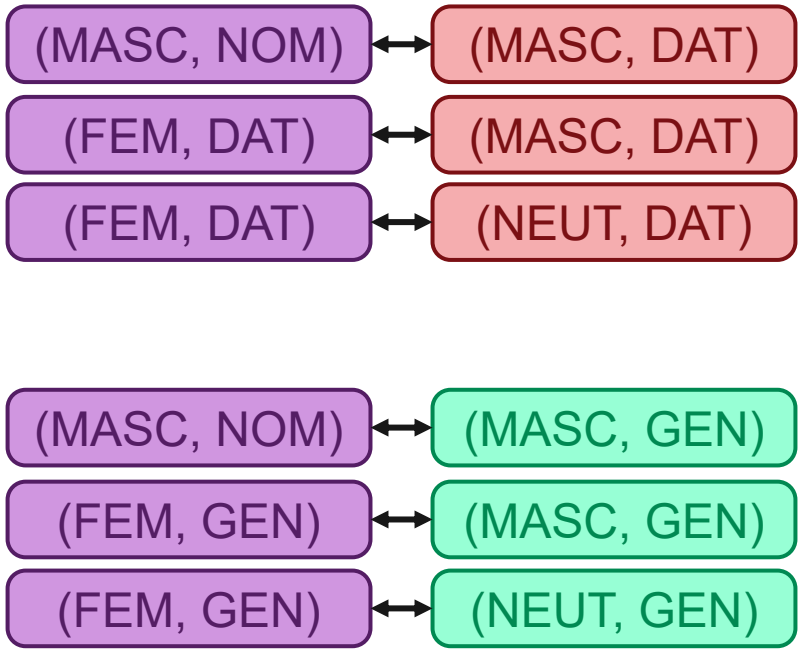
	NOM	ACC	DAT	GEN
MASC	der	den	dem	des
FEM	die	die	der	der
NEUT	das	das	dem	des

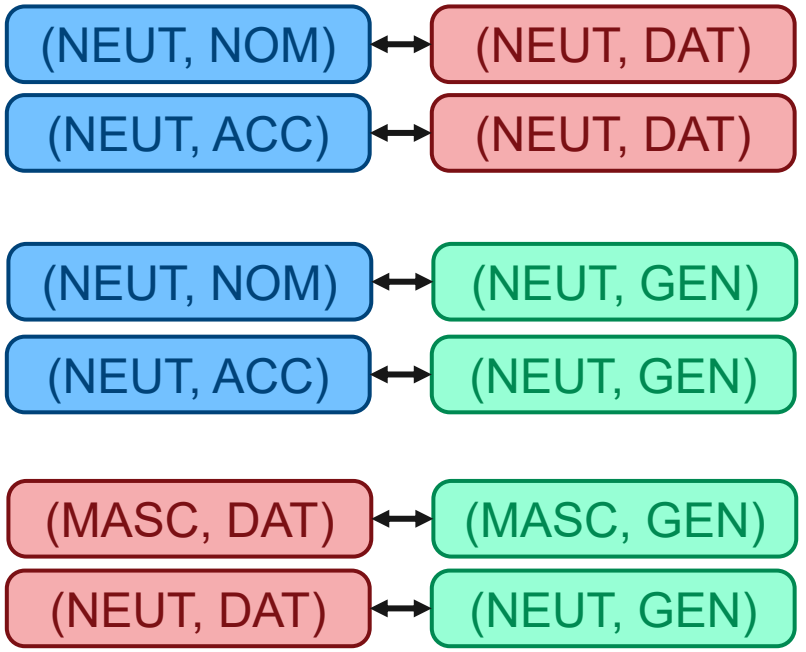
The same article pair can correspond to different gender-case transitions.

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Which Gender-Case Transitions Do We Study?







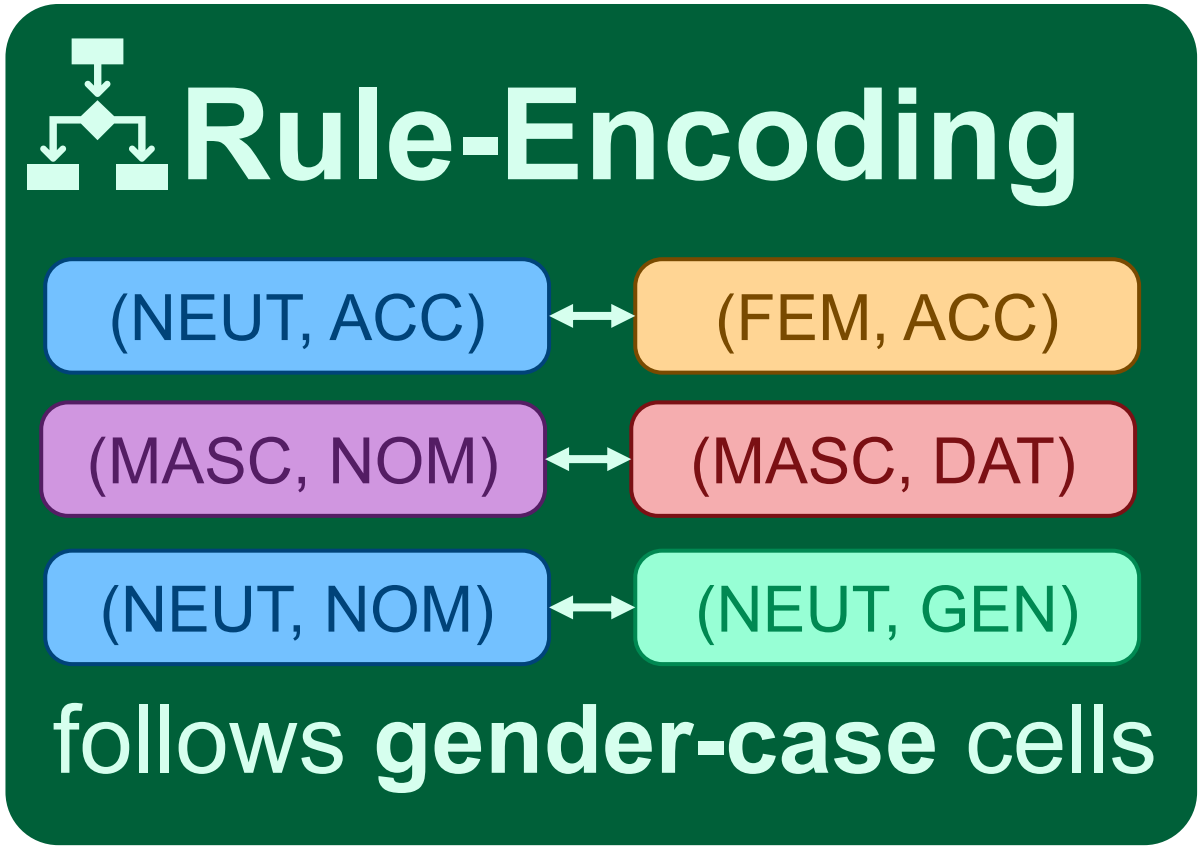
only one dimension changes per transition: gender or case

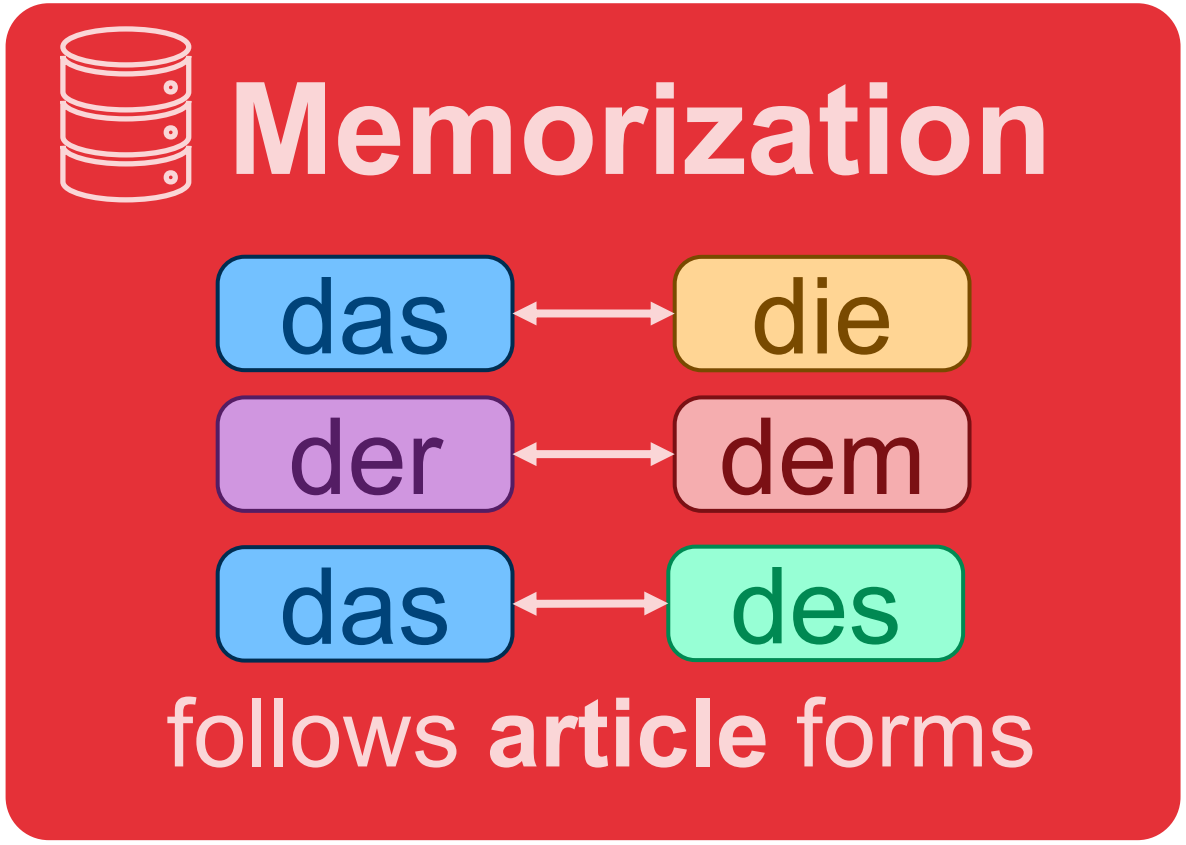
	NOM	ACC	DAT	GEN
MASC	der	den	dem	des
FEM	die	die	der	der
NEUT	das	das	dem	des

19 gender- case cell transitions

Research Question and Hypotheses

Do model mechanisms follow gender–case structure or surface article forms?





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Experimental Setup

Data

German Wikipedia → spaCy gender-case tagging → filtered article contexts → masked gender-case datasets



Models

	Encoder-only	Decoder-only
Monolingual	GBERT, GermanBERT, ModernGBERT	GermanGPT-2
Multilingual	EuroBERT	LLaMA

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GRADIEND: Targeted Feature Learning from Gradients

GRADIEND is a recent interpretabiltiy method [1] learning a single feature neuron of a targeted **gender-case pair**, e.g., (NEUT, ACC) ↔ (FEM, ACC)

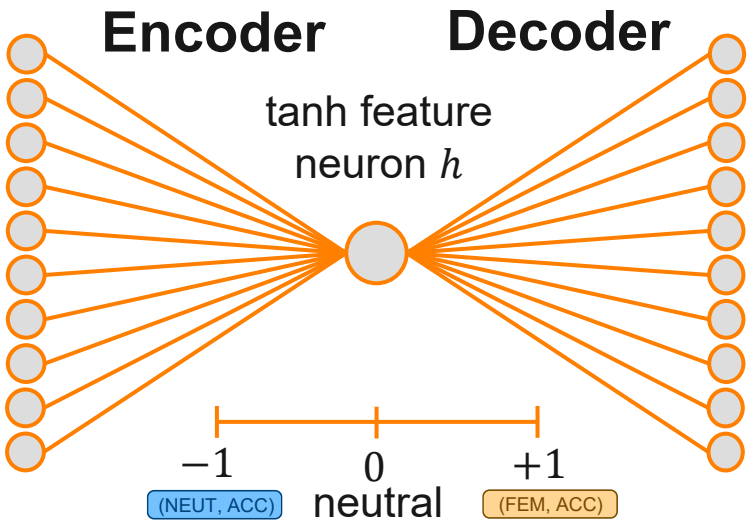
→ one GRADIEND per base LM and transition

Input

Counterfactual gradient over all LM parameters

$$\nabla_c \text{LM}$$

GRADIEND



Output

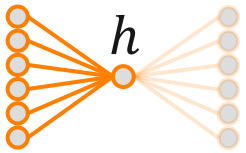
Difference of **Factual** and **Counterfactual** gradient

$$\nabla_c \text{LM} - \nabla_f \text{LM}$$

Takeaway: GRADIEND gives us three objects for analysis – **feature activations**, decoder-based **interventions**, and **affected weights**.

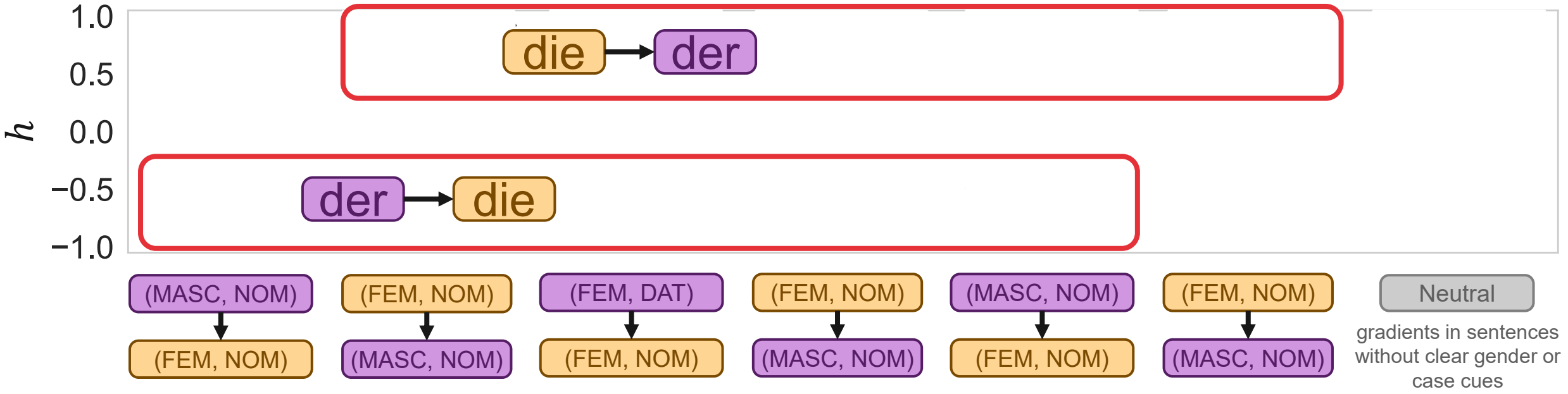
Result 1: Learned Features Are Not Purely Grammatical

How does GRADIEND encode different gradients?



GBERT (MASC, NOM) ↔ (FEM, NOM)

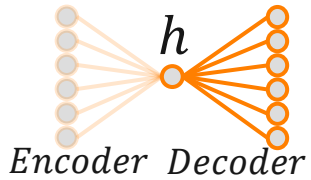
- targeted gender-case pair vs. pairs with same articles



Takeaway: GRADIEND learns a **meaningful** feature, but its encoding also reflects **shared** article forms beyond the targeted gender–case transition.

Result 2: Interventions Show Mixed Patterns

We apply an update to the base model and analyze **probability shifts** on articles.



$$\widetilde{LM} := LM + \alpha \cdot Decoder(h),\ \alpha > 0,\ h := Encoder(\text{(FEM, NOM)})$$

→ shift LM from (MASC, NOM) towards (FEM, NOM)

→ Which article probabilities are affected?

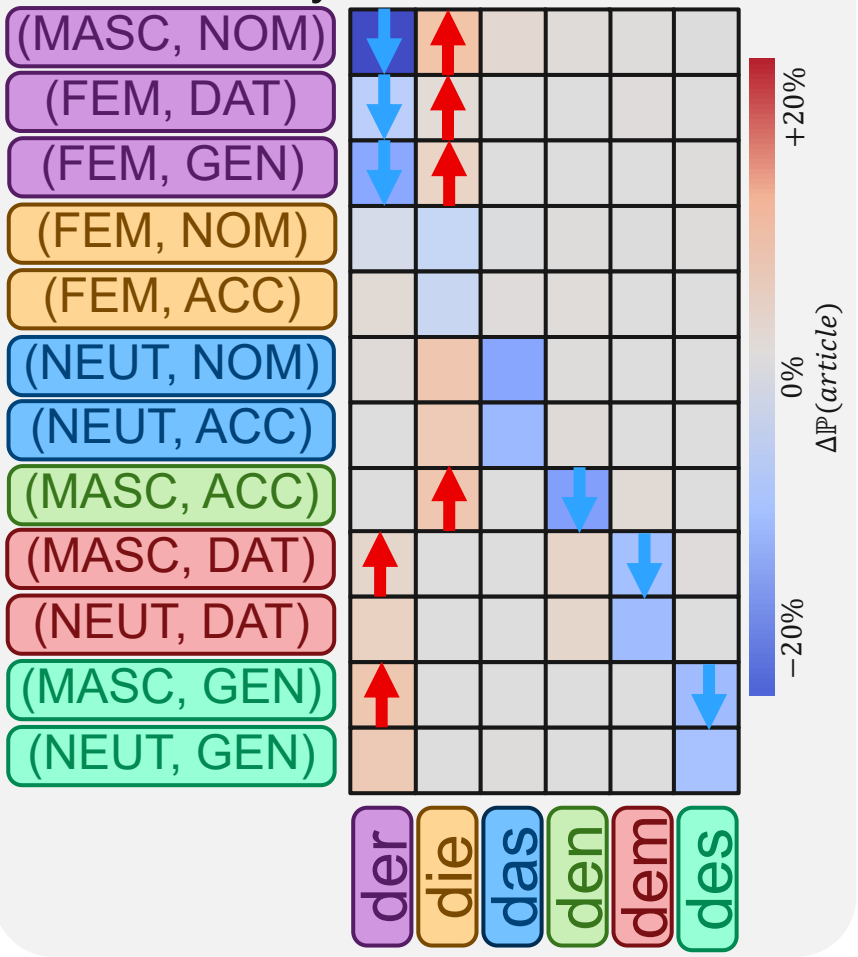
Three tested patterns: **Local Rule** **Generalized Rule** **Spillover**

	NOM	ACC	DAT	GEN
MASC	der	den	dem	des
FEM	die	die	der	der
NEUT	das	das	dem	des

Takeaway: Probability shifts show a **mixed** pattern: some **rule-like** generalization, but also surface-article **spillover**.

GBERT (MASC, NOM) ↔ (FEM, NOM)

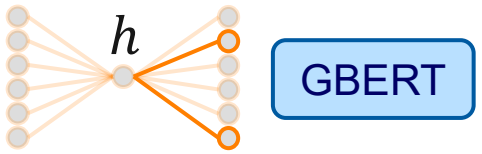
Probability Shift: LM vs. $\widetilde{LM}$



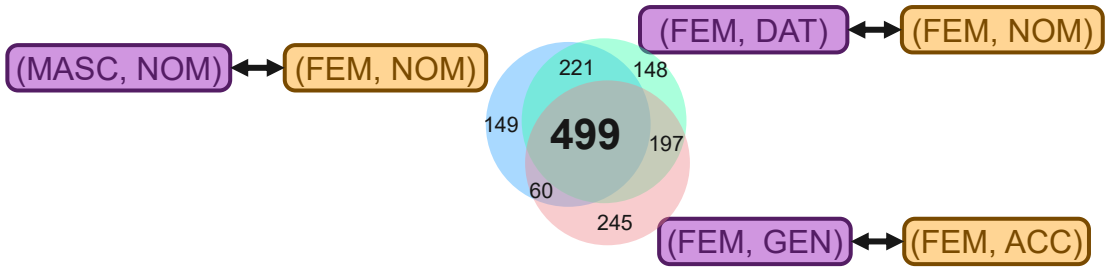
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Result 3: Shared Article Forms Affect Similar Weights

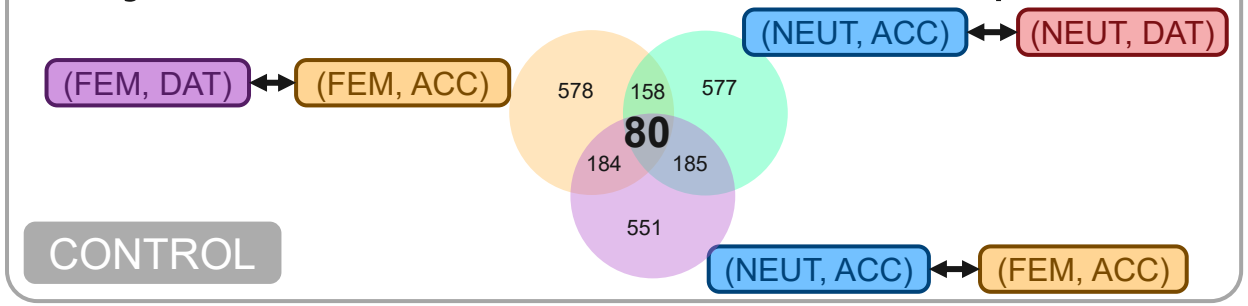
For each GRADIEND, we compare the top-1,000 affected LM weights across transitions.



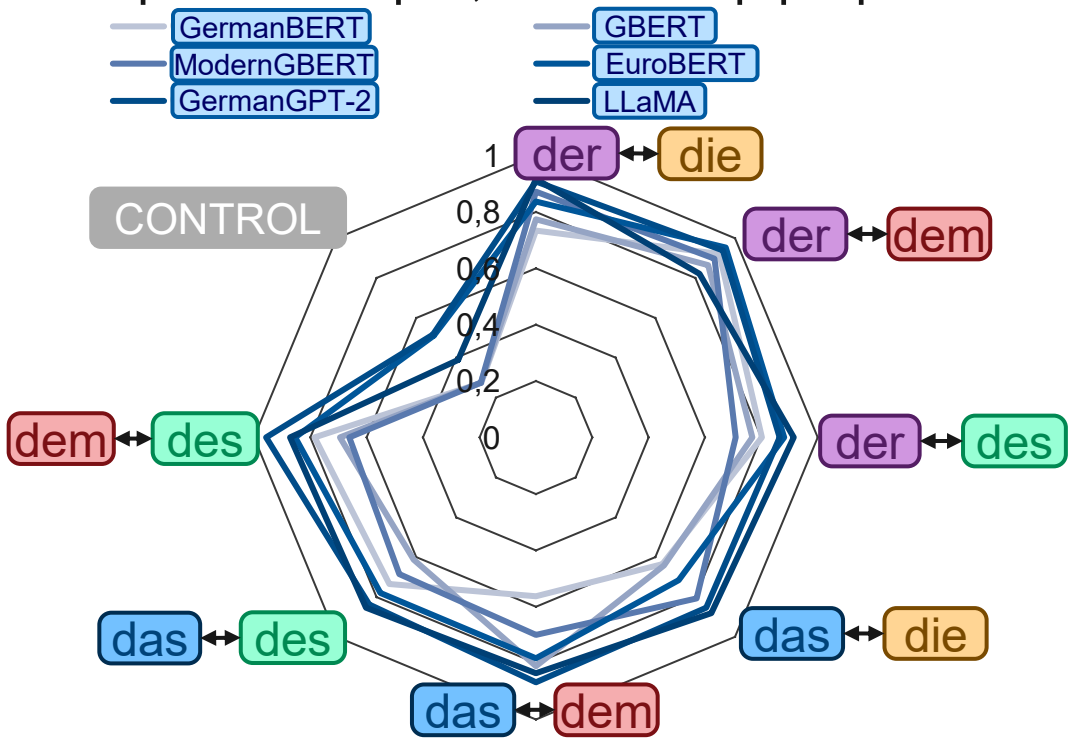
Shared surface articles: **high** overlap



Disjoint/control transitions: **low** overlap



Max. pairwise top-1,000 overlap proportion



Takeaway: Transitions **sharing** surface articles affect more similar weights than **control** transitions.

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Surface-form memorization plays a role

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Not purely rule-based

Understanding or Memorizing?

A Case Study of German Definite Articles in Language Models



WARNING

THE LANGUAGE MODEL YOU ARE USING **MAY HAVE MEMORIZED** THE GRAMMAR **INSTEAD OF LEARNING** THE RULES!

LINGUISTIC RESEARCH RELYING ON HOW LANGUAGE MODELS ENCODE GRAMMAR **MAY BE MISLEADING**.

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